Automating Data Migration of Student Records

Submitted for the partial fulfilment of the requirements for the award of the degree in

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

BY

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DECLARATION

I, **K.C HAARISH (22-UCS-145)** hereby declare that the project report entitled **"AUTOMATING DATA MIGRATION OF STUDENT RECORDS"** is done by me under the guidance of my guide **Dr. S. VENKATALAKSHMI** at LOYOLA COLLEGE (Autonomous), Chennai -34 is be in submitted in partial fulfilment of the requirements for the award of the degree in BACHELOR OF SCIENCE IN COMPUTER SCIENCE.

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BONAFIDE CERTIFICATE

This is to certify that the project work entitled **"AUTOMATING DATA MIGRATION OF STUDENT RECORDS"** is being submitted to Loyola College (Autonomous), Chennai-600034 by **K.C HAARISH (22-UCS-145)** for the partial fulfilment for the award of degree of Bachelor of Computer Science is a bonafide record of work carried out by him, under my guidance and supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

Automating Data Migration of Student Records

ABSTRACT

Data migration is a critical task in many educational institutions, particularly for managing large volumes of student records. The traditional manual transfer of student data from Excel spreadsheets to databases is prone to human error, time-consuming, and inefficient. Robotic Process Automation (RPA), when combined with Python, provides a solution to automate this migration process. This project presents automating data migration of student records from Excel to a database using Python-based RPA tools. The proposed solution leverages Python libraries like Pandas for reading and processing Excel files. The RPA process is designed to handle a variety of Excel formats, validate the accuracy of data, and ensure that the migration is reliable, repeatable, and scalable. Furthermore, the automation script includes error handling, logging, and real-time monitoring to ensure data integrity and minimize downtime during migration. This approach significantly reduces the workload of administrative staff, enhances data accuracy, and accelerates the migration process, ensuring that student records are securely transferred into a centralized database for efficient future use.

FRONTEND: HTML, CSS

BACKEND: PYTHON, MySQL

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